

2024

Bachelor in Business Administration

Course Title: **Business Mathematics II**

Code No: **MTH202** Semester: **2nd Semester**

Full Marks: 60 Pass Marks: 24 Time: period 10 if firm X starts off in time period 0 by setting a price of 300.17. Find the time path of the national income $\{y_t\}$ of the following data and comment on the stability of the time path. $\{y_t = C_t + I_t, I_t = 200 + 0.75y_{t-1}, I_t = 50 + 0.15y_{t-1}\}$ and $\{y_0 = 3000\}$. Section "C" Long Answer Questions: (Attempt any THREE Questions) $[3 \times 5 = 15]$ 18. Solve the following LP problem using the simplex method or graphic method: Maximize $\{Z = 5x_1 + 3x_2\}$. Subject to the constraints $\{2x_1 + x_2 \leq 5\}$ $\{x_1 + x_2 \leq 4\}$ and $\{x_1, x_2 \geq 0\}$ 19. For the following transaction matrix of two sector economy consisting two industries P and Q, calculate the gross output for each industry if the final demand changes to 18 units for P and 44 units for Q. Producer User Final demand Total output P Q P1620440 Q840328020. Solve the difference equation: $\{y_t = 0.6y_{t-1} + 21, y_0 = 50\}$. Find $\{y_5\}$ and $\{y_{11}\}$? 21. In a competitive market price where $\{Q_d = 500 - 5P\}$ and $\{Q_s = 40 + 20P\}$, the initial price $\{P(0)\}$ is Rs 50. (a) Derive a function for the time-path of P and use it to predict price in time period 5 given that price adjusts in proportion to excess demand at the rate $\{\frac{dP}{dt} = 0.01 (Q_d - Q_s)\}$. (b) How many time periods would you have to wait for the price to drop by Rs. 30? Section "D" Comprehensive Answer / Case / Situation Analysis Questions: $[2 \times 10 = 20]$ 22. The total cost and demand functions for two goods X and Y are given by the equations $\{TC = 10 - 6x + 5y\}$ $\{P_x = 6 - 2x\}$ $\{P_y = 25 - 0.1y\}$ where x and y are the number of units of the first goods and second goods respectively. a. Write down the equations for total revenue and total profit. b. Determine the number of units of each good which should be sold to maximize revenue. Calculate the maximum revenue and prices of each good under revenue maximizations. c. Determine the number of units of each good which should be bought and sold to maximize profits. d. Calculate the maximum profit, prices of each good and total revenue under profit maximization.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions: $[10 \times 1 = 10]$

1. Integrate: $\int \frac{x+2}{x-5} dx$
2. Find the area of the region bounded by the curve $\{y = \frac{3}{2}x^2\}$, the x-axis and the two ordinates $\{x = 0\}$ and $\{x = 2\}$.
3. Find $\{f_{xx}\}$ when $\{f(x, y) = 3x^2y^2 + 5x^2y\}$.
4. If $\{A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}\}$, find $\{A^2\}$.
5. Evaluate: $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$
6. The input-output coefficient matrix of an economy of two industries is $\begin{bmatrix} 0.8 & 0.2 \\ 0.9 & 0.7 \end{bmatrix}$. Test whether the system is viable as per Hawkins-Simon conditions.
7. Determine the order and degree of the differential equations: $\left(\frac{dy}{dx}\right)^2 - \left(\frac{dy}{dx}\right)^5 + 7xy = 0$.
8. Solve the differential equation: $\frac{dy}{dx} = \frac{1}{x}$

9. Solve the first order homogeneous difference equations: $(y_{t+1} - 2y_t = 0, y_0 = 3)$
10. The marginal cost function of manufacturing (x) units of a product is given by $MC = (3x^2 - 10x + 3)$. The total cost of producing one unit of the product is Rs. 7. Find average cost function.

Section B

Short Answer Questions: (Attempt any FIVE Questions) $[5 \times 3 = 15]$

11. Integrate the following: (a) $\int \frac{\log x + 5}{x} dx$ (b) $\int (2x + 1)\sqrt{5x + 2} dx$
12. The demand function for a product is $(P_d = 8 - 2x)$ and the supply function is $(P_s = 2x)$ under pure competition. Find the consumer's surplus and producer's surplus.
13. Solve the following system of linear equations using Cramer's rule or matrix method: $(4x + 6z = 100)$ $(3x + 6y + z = 100)$ $(3x + 4y + 3z = 100)$
14. A firm's production function is given by the equation $(Q = 100L^{0.5}K^{0.5})$. Use partial differentiation to find the approximate change in output (Q) when the firm's manager increases capital by 5% and decreases labour by 3%.
15. Solve the differential equations: $[\frac{dy}{dx} + \frac{1}{x}] \cdot y = x^3$
16. The reaction functions for the two duopolists, firms X and Y are given by the equations $(P_t^X = 45 + 0.8 P_{t-1}^Y)$ and $(P_t^Y = 45 + 0.8 P_{t-1}^X)$, respectively. If the assumptions of the Bertrand model hold, derive a difference equation for (P_t^X) and calculate what (P_t^X) will be in time period 10 if firm X starts off in time period 0 by setting a price of 300.
17. Find the time path of the national income (y_t) of the following data and comment on the stability of the time path. $(y_t = C_t + I_t)$ $(C_t = 200 + 0.75y_{t-1})$ $(I_t = 50 + 0.15y_{t-1})$ and $(y_0 = 3000)$.

Section C

Long Answer Questions: (Attempt any THREE Questions) $[3 \times 5 = 15]$

18. Solve the following LP problem using the simplex method or graphic method: Maximize $(Z = 5x_1 + 3x_2)$. Subject to the constraints $(2x_1 + x_2 \leq 5)$ $(x_1 + x_2 \leq 4)$ and $(x_1, x_2 \geq 0)$
19. For the following transaction matrix of two sector economy consisting two industries P and Q, calculate the gross output for each industry if the final demand changes to 18 units for P and 44 units for Q.
- | | Producer | User | Final demand | Total output |
|---|----------|------|--------------|--------------|
| P | 16 | 20 | 44 | 80 |
| Q | 8 | 40 | 32 | 80 |
20. Solve the difference equation: $(y_t = 0.6y_{t-1} + 21, y_0 = 50)$. Find (y_5) and (y_1) ?
21. In a competitive market price where $(Q_d = 500 - 5P)$ and $(Q_s = 40 + 20P)$, the initial price $(P(0))$ is Rs 50. (a) Derive a function for the time-path of P and use it to predict price in time period 5 given that price adjusts in proportion to excess demand at the rate $(\frac{dP}{dt} = 0.01 (Q_d - Q_s))$. (b) How many time periods would you have to wait for the price to drop by Rs. 30?

Section D

Comprehensive Answer / Case / Situation Analysis Questions: $[2 \times 10 = 20]$

22. The total cost and demand functions for two goods X and Y are given by the equations $(TC = 10 - 6x + 5y)$ $(P_x = 6 - 2x)$ $(P_y = 25 - 0.1y)$ where x and y are the number of units of the first goods and second goods

respectively. a. Write down the equations for total revenue and total profit. b. Determine the number of units of each good which should be sold to maximize revenue. Calculate the maximum revenue and prices of each good under revenue maximization. c. Determine the number of units of each good which should be bought and sold to maximize profits. d. Calculate the maximum profit, prices of each good and total revenue under profit maximization.